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The toxicity of microplastics and their leachates to embryonic development of the sea cucumber *Apostichopus japonicus*

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ABSTRACT

Microplastic pollution has been widely detected across the global ocean, posing a major threat to a wide variety of marine biota. To date, the deleterious impacts of microplastics have predominantly been linked to their direct exposure, while the potential risks posed by the leachates emanating from microplastics have received comparatively less attention. Here, the toxicity of virgin plasticized polyvinyl chloride (PVC) microspheres and their leachates were evaluated on the embryo-larval development of sea cucumber *Apostichopus japonicus* using an in-vitro assay. Results showed that a significant toxic effect of both PVC microspheres and their leachates on the embryo development and larval growth of sea cucumbers follows a dose-dependent and time-dependent pattern. Nonetheless, the toxicity of PVC leachates surpasses that of the microspheres themselves. Abnormal developmental phenotypes, such as aberrant gastrulation, misaligned mesenchymal cells, and delayed arm development, were also observed in embryos and larvae treated with PVC. Further chemical analyses of PVC microspheres and leachates revealed the existence of five distinct phthalate esters (PAEs), with DIBP (diisobutyl phthalate) and DBP (dibutyl phthalate) exhibiting higher concentrations in the PVC leachates. This finding suggests that the elevated toxicity of plastic leachate may be attributed to the leaching of phthalate additives from the plastic particles.

1. Introduction

Plastic litters have been detected across global marine ecosystems (Andrady, 2011). According to estimates, approximately 10% of the annual production of 300 million tonnes of plastic finds its way into the ocean, thereby constituting a significant proportion of marine debris, ranging from 60% to 80% (Derraik, 2002; Ritchie and Roser, 2018). In recent years, microplastics (particles less than 5 mm, MPs) as an emerging environmental pollutant, have received worldwide concern (Wright et al., 2013; Moulin and van Egmond, 2019). Due to their ubiquity and persistence, MPs have been recognized as a global threat to a wide range of marine wildlife (Derraik, 2002; Browne et al., 2008; Beiras et al., 2018; Wright et al., 2013; Mohsen et al., 2021). Accumulating evidence has revealed that ingestion of MPs can have detrimental effects on marine organisms, including reducing feeding activity, physical damage of digestive system, retarding growth and development, inducing oxidative stress, energy metabolism disorder, immunotoxicity and even microbiota dysbiosis (Murray and Cowie, 2011; Cole et al., 2015; Revel et al., 2018; Song et al., 2019; Teng et al., 2021; Zhu et al., 2022).

Compared to the direct impact of MPs particles exposure, less attention has been paid to potential hazards associated with the additive chemicals leached out from MPs (Delaeter et al., 2022). Plasticizers are crucial additives used in industrial plastics processing, particularly in polyvinyl chloride (PVC)-based products, to enhance polymer performance. However, these plasticizers are easily leached from plastic materials into water (Net et al., 2015; Hahladakis et al., 2018). By far, most studies about the toxicity of plastic leachates have mainly been conducted on model animals. For example, toxic effects of plastic leachates have been evaluated for copepodes (Lithner et al., 2012; Bejgarn et al., 2015), barnacle larvae (Li et al., 2015), mussel embryos (Gandara e Silva et al., 2016; Magni et al., 2018) and sea urchin embryos (Nobre et al., 2015; Oliviero et al., 2019). Results from these studies have shown that plastic leachates can pose an impact on the health and development of

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marine biota, but the severity of this impact varies with different plastic items and distinct species (Hermabessiere et al., 2017). Hence, further research is required to validate the potential effects of plastic leachates on a broader range of marine species.

Sea cucumber *Apostichopus japonicus* is one of the most valuable mariculture species in Asian countries, especially in China. By 2021, the mariculture production of this species has reached 222,707 tons in China, with the mariculture areas covering 24 thousand hectares (Ministry of Agriculture and Rural Affairs of China, 2022). Meanwhile, as a typical benthic sedimental feeder, this species also plays an important ecological function in marine ecosystems through organic matter removing and nutrient cycling (Michio et al., 2002; MacTavish et al., 2012). Given the benthic environment will become the final sink of MPs (Oliveira and Almeidam, 2019; Peng et al., 2020), macro-invertebrates with a benthic lifestyle, e.g. *A. japonicus*, are easily affected by the MPs pollution (Candia Carnevali, 2005). Previous studies have indicated that plastic fragments could be ingested and transferred by adult sea cucumbers to their coelomic fluid along with sediment (Graham and Thompson, 2009; Mohsen et al., 2019, 2020a, 2020b). Like the vast majority of marine species, *A. japonicus* has a pelagic larval stage, relying on small particles similar in size to MPs as food (Kaposi et al., 2014). Moreover, the embryo and auricularia stages are widely acknowledged as the most susceptible to environmental fluctuations in the life cycle of sea cucumbers (Li and Li, 2010; Li et al., 2011), which can have consequential ramifications for individual survival and population dynamics. However, the toxic effects of MPs on sea cucumber embryos and larvae still remain poorly understood. Therefore, it is important and imperative to explore the impacts of MP exposure on early life history of *A. japonicus* to clarify the potential effects of MP pollution on the breeding and population recruitment of this species.

The aim of this study was to examine the toxic effects of MPs (PVC, Φ10 μm) and their leachates on embryonic development of the sea cucumber *A. japonicus*. Additionally, gas chromatography analysis was performed in order to better understand the toxicity in relation to plastic additives. Thus, this study seeks to contribute new understanding of both the potential effects of MPs and chemicals leached form the plastic on the *A. japonicus* embryos and larvae.

2. Materials and methods

2.1. Materials and animals

2.1.1. Preparation of microplastic suspensions

The virgin white PVC microspheres (PVC-microspheres) with a size of 10 μm were purchased from Shanghai Yangli Electromechanical Technology Co., Ltd., China. The PVC suspensions were prepared by weighing and adding the PVC microspheres to sterile seawater to obtain a stock solution of 90 mg L^{-1}. For better dispersion of PVC microspheres, the suspensions were sonicated for 30 min plus surfactant Tween 20 before use (Lu et al., 2016). Previous studies and our pre-experimental results revealed that Tween 20 at a concentration (3 μl L^{-1}) did not affect sea cucumber embryonic and larval development (Beiras et al., 2018).

2.1.2. Preparation of microplastic leachates

Methods to prepare the PVC leachates (PVC-leachates) draws on the approach from Nobre and collaborators (Nobre et al., 2015). Briefly, the PVC microspheres were firstly suspended at a dilution concentration of 0.3, 1, 3, and 10 mg L^{-1} plus 3 μl L^{-1} Tween 20 as mentioned in section 2.1.1., and then the microspheres were leached on a constant-temperature oscillator from Suzhou Peiying Experimental Equipment Co., LTD (Heidolph Unimax, 2010) at 28 °C with shaking rate at 150 rpm for 72 h in the dark. Next, PVC leachates were obtained by filtering through Whatman GF/C glass microfiber membrane filters from Shanghai Xingya purification Materials Factory (50 mm in diameter, pore size 8 μm) to remove particles. The microplastic microspheres were removed from the leachates to serve two purposes: first, to avoid potential mechanical effects of particles on the sea cucumber embryos by direct interaction; second, to assure leachates at uniform concentrations during exposure experiments (Rendell-Bhatti et al., 2021).

2.1.3. Sea cucumber embryos and larvae

Mature male and female sea cucumbers (*A. japonicus*) were collected in April and May 2022 (reproductive period of the species) from the culture farm located in Yantai (Shandong Province, China). After transferring to the laboratory and acclimating for a week (16 °C, 29–31 PSU, 7 mg L^{-1} DO and 12 h light: 12 h dark), the adults were then introduced to spawn using the air-drying method. The oocytes and sperms were harvested and examined for maturity by checking sperms for motility and eggs for roundness using a light microscope (zoom × 100). Sperms were transferred slowly to the mature oocytes in a 50 mL glass beaker, and gentle shaking to promote fertilization. Fertilization success was assessed to ensure fertilization rate is higher than 95%. Developing embryos were kept in a dark incubator at 22 ± 1 °C, and reached two major developmental timepoints at gastrula stage (24 h post fertilization, 24 hpf) and early auricularia stage (48 h post fertilization, 48 hpf) in normal conditions.

2.2. Embryotoxicity bioassays

Toxicity of microplastics and leachates were evaluated by assessing the embryo development and larvae survival. Microplastics exposure concentrations were chosen on the basis of trial experiments and previous studies by dilution in seawater plus Tween 20 (3 μl L^{-1}) at following concentrations: 0.3, 1, 3, and 10 mg L^{-1} (reaching the lowest concentration to elicit effects in sea cucumber embryos) (Oliviero et al., 2019). Test leachates dilutions were prepared in section 2.1.2. At 10 min post-fertilization, 1 mL aliquot of the healthy fertilized eggs (approximately 50 eggs mL^{-1}) was added to 9 mL PVC suspensions or leachates, at testing concentrations of 0.3, 1, 3, and 10 mg L^{-1} in a six-well tissue culture plate. In order to reduce the possibility of phthalate relase in PVC microsphere exposure experiments, the exposure medium was changed daily. For each assay, three trials were conducted with three replicates for each treatment, while non-treated negative control samples in only seawater plus Tween 20 (3 μl L^{-1}) were run simultaneously thrice in 3 replicates. Embryos were thus reared in a dark incubator at 22 ± 1 °C.

2.3. Endpoints

2.3.1. Survival index

The embryos in each treatment were sampled at 24 hpf and 48 hpf and immediately fixed with 40% formaldehyde respectively, and the percentages of dead, deformed, and hatched numbers were observed with a Stereomicroscope (6.4–40× magnification, SZX-10, Olympus, Japan) to calculate the mortality (dead number/total number of embryos × 100%), teratogenicity rate (impaired number/total number of embryos × 100%), and hatchability (hatched number/total number of embryos × 100%). The half-effect concentration (EC_{50}) and half-lethal concentration (LC_{50}) of PVC suspensions or leachates on embryos is defined as causing 50% of embryos to develop abnormally and to die, respectively. LC_{50} and EC_{50} values and their 95% confidence intervals were calculated by the Probit dose-response model. To assess the teratogenic potential of a test substance, the calculation of the teratogenicity index (TI) is traditionally employed as the quotient of LC_{50} and EC_{50}. If the TI of a given substance is > 1, the substance is considered to be teratogenic; if the TI is ≤ 1, the substance produces mainly embryolethal effects (Weigt et al., 2012). Upon discovering significant differences ($p < 0.05$) among groups through ANOVA, Dunnett's post hoc test was employed to determine the minimum no observed adverse effects concentration (NOEC) and the minimum observed adverse effect concentration (LOEC) for each treatment in comparison to the control.

2.3.2. Developmental index

To assess the effects of PVC microspheres and leachates on development of sea cucumber embryos and larvae, 81 randomly chosen fixed individuals of each tested concentration group were photographed under an Stereomicroscope (6.4–40× magnification, SZX-10, Olympus, Japan). The body width (BW), body length (BL), horizontal stomach diameter (S1), and vertical stomach diameter (S2) (Fig. 1) were measured to verify the presence of developmental delay or malformations as adapted from Beiras et al. (2012) and Zhan et al. (2019). In addition, superficial areas of *A. japonicus* embryos and larvae were measured using PhotoShop 2022 software with the magic wand tool. Stomach volume (SV) was further calculated with the following formula:

$$SV = 4/3\pi \times [(S1+S2)/4]^{3}$$

To quantitatively evaluate the developmental toxicity of PVC microspheres and leachates exposure, sea cucumber embryos and larvae were classified into four categories according to a previous study of Rendell-Bhatti et al. (2021) based on morphological deviation from the normal developed individuals (Table 1 and Fig. 2). Furthermore, in order to determine the frequency of each category of embryo or larval alteration, an index of microplastic impact (IMPI) was calculated with the following formula:

$$IMPI = [0 \times \%\ category0+1 \times \%\ category1+2 \times \%\ category2+3 \times \%\ category3]/100$$

The IMPI scale provides a continuous spectrum ranging from 0, indicating no impact, to 3, indicating complete mortality. Intermediate levels correspond to varying degrees of developmental delays and morphological malformations, as previously outlined (Table 1).

2.4. Microplastic suspensions and leachate chemistry analysis

According to the criteria for determination of phthalate acid easters (PAEs) in marine environment (HY/T 170–2015, China), the phthalate plasticizers were analyzed in the original PVC suspension and PVC leachate (10 mg L^{-1}) by gas chromatagraphy-mass spectrometry. Ultrasonic extraction of n-hexane was carried out by gas chromatography-mass spectrometer (QDCH-E596). The sample was injected at 300 °C in constant flow mode with a volume of 1.0 μL. The separation was performed in a medium polar capillary column (35% fillers were phenylmethylpolysiloxane) with the specifications (length × inner diameter × film thickness of 30.0 m × 0.25 mm × 0.25 μm). The chromatographic conditions were as follows: flow rate was 1.2 mL min^{-1}, initial column temperature was maintained at 100 °C for 0.5 min, and temperature was raised at 25 °C min^{-1} to 300 °C for 2 min. Carrier gas: high purity helium (purity ≥99.999%). Ion source: electron bombardment ion source (EI); Chromatography-mass spectrometry interface temperature 290 °C; Ion source temperature 280 °C; Quadrupole temperature 150 °C; Solvent delay 3.8 min; Determination method: Select ion monitoring (SIM); Ionization energy: 70 eV. Ion detection mode was selected, retention time and relative abundance of characteristic ions were qualitatively analyzed, sample solution and standard solution were selected, and peak area was used for quantitative analysis.

Table 1
Morphological classification of sea cucumber embryos and larvae.

24 hpf embryo classification	
Category	Characteristic features
0	Gastrula stage, interstitial cell formed, well-structured, typical archenteron, the size is N
1	The embryo's endoderm is developmentally delayed and does not reach half of the embryo body, but with clear borders, size <2/3N; the mesenchymal cells random distribution
2	Mesenchymal blastocyst with irregular morphology
3	Highly abnormal features, no clear borders, necrosis
48 hpf larva classification	
Category	Characteristic features
0	Digestive tract completed, gastric enlargement, typical early auricularia stage larval morphology
1	Delayed development, irregular shape, unclear arms
2	Mesenchymal blastocyst
3	Highly abnormal features, no complete clear border, necrosis

2.5. Statistical analysis

Data was given as mean ± standard deviation. One-way analysis of variance (ANOVA) by using SPSS Statistics 21.0 software was used for the comparison between groups with different concentrations, and paired-comparison *t*-test was used for the comparison between groups with the same concentration in PVC-microspheres and PVC-leachates groups. Significance was indicated at $p < 0.05$.

3. Results

3.1. Survival Index

3.1.1. Mortality, teratogenicity and hatchability

As illustrated in Fig. 3A and Fig. 3. A′, both PVC microsphere and

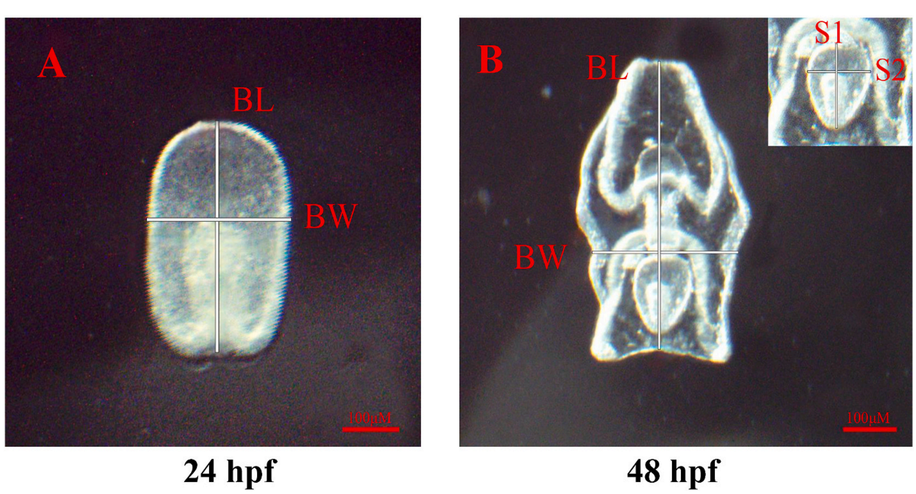

Fig. 1. Embryo and larvae of sea cucumber *A. japonicus*.
A. Normal sea cucumber embryo at 24 hpf. B. Normal sea cucumber larvae at 48 hpf. BW: body width. BL: body length. S1: horizontal diameter of stomach. S2: vertical diameter of stomach. Images × 40 magnification, scale bar: 100 μm.

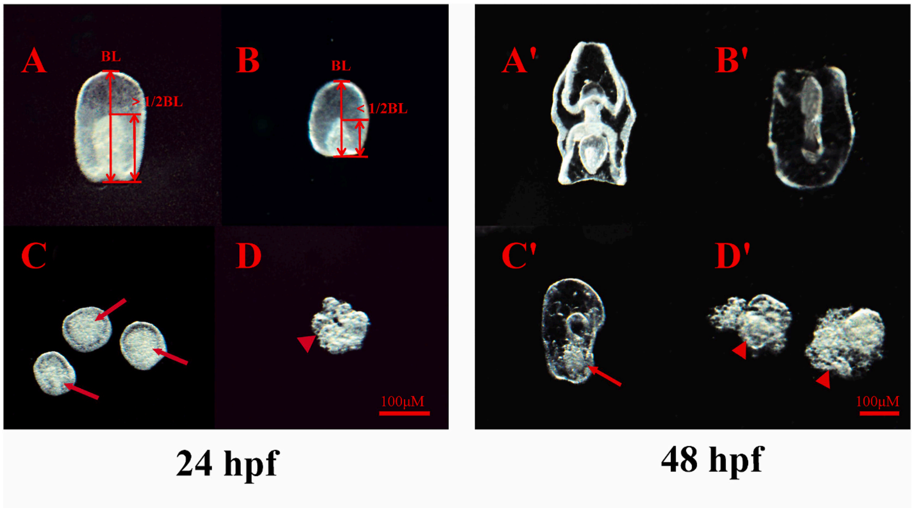

Fig. 2. Micrographs of four morphological categories in sea cucumber embryos or larvae.
A, B, C, and D represents the corresponding phenotypes of embryos (24 hpf) at category 0, 1, 2 and 3, respectively. A′, B′, C′, and D′ represents the corresponding phenotypes of larvae (48 hpf) at category 0, 1, 2 and 3. Red arrows refer to mesenchymal blastocyst with irregular morphology. Red triangle refer to unclear borders, necrosis. Images × 40 magnification, scale bar: 100 μm.

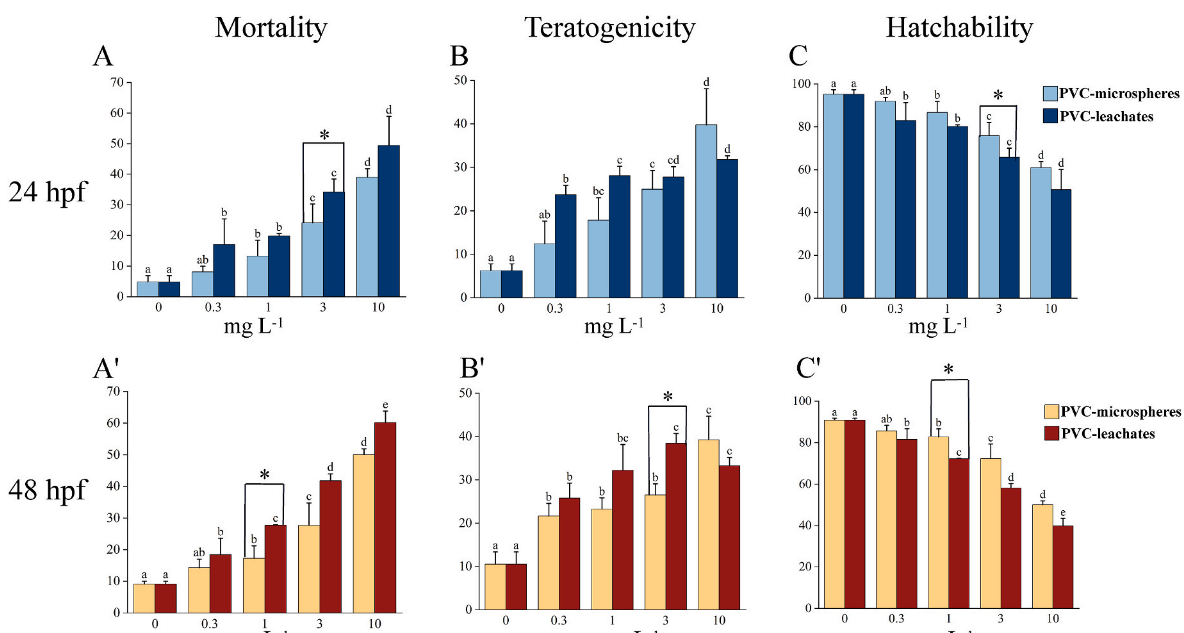

Fig. 3. Growth and survival analysis of *A. japonicus* embryos and larvae exposed to PVC.
Different letters indicate statistical differences ($p < 0.05$) between different concentrations group using ANOVA analysis. An asterisk (*) indicate statistical differences ($p < 0.05$) between groups with the same concentration in PVC-microspheres and PVC-leachates groups using paired-comparison *t*-test.

leachate concentration showed a significant effect on the mortality of sea cucumber embryos after 24 h and 48 h exposure ($p < 0.05$). Highest mortality (49.40% and 60.18%) was observed in highest concentration of PVC-leachates group at 10 mg L^{-1} at 24 hpf and 48 hpf, respectively. The PVC leachates showed more severe toxic effects on the embryos than the PVC microsphere themselves in each concentration, and paired *t*-test revealed that a significant difference in mortality for PVC-leachates group vs. PVC-microspheres group at 3 mg L^{-1} and 1 mg L^{-1}, respectively ($p < 0.05$). An elevation of teratogenicity rate was also observed as PVC microsphere and leachate concentration increases ($p < 0.05$) (Fig. 3B and Fig. 3. B′). The teratogenicity rate in PVC-leachates group was higher than in PVC-microspheres group at concentrations below 3 mg L^{-1}, but the situation is opposite at 10 mg L^{-1}. The 3 mg L^{-1} of PVC-microspheres and PVC-leachates exposure led to a significant difference of teratogenicity at 48 hfp ($p < 0.05$). Percentages of hatched embryos were also significantly affected by PVC microsphere and leachate ($p < 0.05$) (Fig. 3C and Fig. 3. C’), and the 10 mg L^{-1} PVC-leachates microplastic group only had a hatchability rate of 39.82% in *A. japonicus* embryos at 48 hfp. Moreover, the PVC leachates exposure led to more unhatched *A. japonicus* embryos than the microspheres themselves, and the significant differences were observed also at 3 mg L^{-1} and 1 mg L^{-1} at 24 hfp and 48 hfp, respectively ($p < 0.05$).

3.1.2. NOEC, LOEC, LC_{50}, and EC_{50} calculation

The NOEC, LOEC, LC_{50}, EC_{50} and TI values were summarized both in PVC-microspheres and PVC-leachates groups (Table 2). The NOEC and LOEC values of PVC-microspheres group at 24 hpf were 3 mg L^{-1} and 10 mg L^{-1}, and the values of other groups are all equal or less than 0.3 mg L^{-1}. The LC_{50} values of PVC-microspheres group at 24 hpf and 48 hpf were 27.78 mg L^{-1} and 13.79 mg L^{-1}, while the LC_{50} values of PVC-leachates group were 13.22 mg L^{-1} and 5.21 mg L^{-1}, accounting for 47.59% and 37.78% the former PVC-microspheres group values. The EC_{50} values of PVC-leachates group at 24 hpf and 48 hpf were 30.44 mg L^{-1} and 18.27 mg L^{-1}, and the TI values of PVC-leachates group at 24

Table 2
The NOEC, LOEC, LC_{50}, EC_{50} values and their 95% confidence interval for the PVC microspheres and leachates.

Stages	24 hpf		48 hpf	
Treatment	PVC-microspheres	PVC-leachates	PVC-microspheres	PVC-leachates
NOEC(mg L^{-1})	3	<0.3	<0.3	<0.3
LOEC(mg L^{-1})	10	<0.3	0.3	<0.3
LC_{50} (mg L^{-1})	27.78	13.22	13.79	5.21
95% confidence interval	9.843–607.37	5.40–179.89	6.29–89.33	2.84–15.71
EC_{50} (mg L^{-1})	47.83	30.44	16.99	18.27
95% confidence interval	28.58–134.97	18.04–87.41	8.50–75.70	10.81–74.27
TI	0.58	0.43	0.81	0.29

Note: PVC-microspheres represents microplastic suspensions, PVC-leachates represents microplastic leachates.

hpf and 48 hpf were 0.43 and 0.29, respectively.

3.2. Embryo-larval development

3.2.1. Apparent developmental parameters

The effects of PVC microspheres and leachates on larval growth (body shape index and digestive tract index) of *A. japonicus* were shown in Table 3. A significant effect of both PVC microsphere and leachate concentration on body length, body width and size of sea cucumber embryos was seen at the present study ($p < 0.05$). The embryos exposed to 0.3 mg L^{-1} of PVC-microspheres were able to develop normally, but the BL, BW and size of sea cucumber embryos significantly decreased at the groups above 1 mg L^{-1} of PVC-microspheres ($p < 0.05$). Moreover, the severity of developmental delay was found to be directly correlated to the concentration of plastics and exposing time. At the highest tested concentration of PVC-microspheres (10 mg L^{-1}), the BL, BW and size of embryos represent a percentage of 75.40%, 83.10%, 55.65% (at 24 hpf) and 71.55%, 83.64% and 50.54% (at 48 hpf) of the normal developed individuals. Yet exposure to PVC leachates indicated a significant developmental delay starting from the low concentration of 0.3 mg L^{-1} ($p < 0.05$), and a concentration-dependent and time-dependent toxicity were also recorded in PVC-leachates exposure groups. In 10 mg L^{-1} PVC-leachates group, the BL, BW and size of embryos represent a percentage

Table 3
Development index of *A. japonicus* embryos and larvae exposure to PVC microspheres and leachates.

Stage	Treatment (mg L^{-1})		Body Shape Index			Digestive Tract Index			Rating Index
			BL(μm)	BW(μm)	Size(μm^2)	S1(μm)	S2(μm)	SV(μm^3)	IMPI
24 hpf	**PVC microspheres**	0	245.84 ± 9.69^a	175.49 ± 4.75^a	35064.32 ± 1641.06^a	–	–	–	44.14^a
		0.3	241.48 ± 5.59^{ab}	170.70 ± 2.96^{ab}	32413.93 ± 4002.86^a	–	–	–	55.82^a
		1	230.40 ± 5.80^b	166.19 ± 44.63^b	26740.44 ± 3673.97^b	–	–	–	100.80^b
		3	202.92 ± 9.50^c	154.33 ± 2.50^c	22111.16 ± 2393.31^c	–	–	–	141.27^c
		10	185.37 ± 7.02^d	145.84 ± 3.99^d	19512.15 ± 2751.68^c	–	–	–	178.13^d
	PVC leachates	0	245.84 ± 9.69^a	175.49 ± 4.75^a	35064.32 ± 1641.06^a	–	–	–	44.14^a
		0.3	220.62 ± 3.88^b	170.39 ± 1.96^a	31025.40 ± 3541.04^b	–	–	–	98.53^b
		1	197.97 ± 6.99^c	161.63 ± 3.84^b	25528.32 ± 2572.69^c	–	–	–	128.07^b
		3	185.77 ± 6.19^d	153.96 ± 4.39^c	21552.14 ± 3162.17^d	–	–	–	166.87^c
		10	166.73 ± 2.51^e	146.84 ± 7.20^d	17876.14 ± 1020.49^d	–	–	–	219.60^d
48 hpf	**PVC microspheres**	0	384.05 ± 4.95^a	197.18 ± 5.91^a	98221.63 ± 2188.37^a	121.28 ± 4.43^a	77.02 ± 3.17^a	510756.67 ± 28826.14^a	53.64^a
		0.3	378.37 ± 3.63^a	192.96 ± 3.65^a	91765.85 ± 4926.26^a	118.88 ± 3.51^{ab}	74.36 ± 4.05^{ab}	474011.30 ± 54021.72^a	74.42^a
		1	356.84 ± 13.66^b	180.91 ± 3.29^b	79324.70 ± 8246.67^b	112.70 ± 5.35^b	69.49 ± 6.15^b	397765.24 ± 54082.60^b	111.50^b
		3	293.72 ± 11.94^c	173.41 ± 4.83^c	66349.74 ± 1976.46^c	93.44 ± 6.81^c	57.69 ± 3.60^c	228257.12 ± 43482.55^c	159.00^c
		10	274.78 ± 6.53^d	164.93 ± 4.49^d	49639.07 ± 2297.94^d	81.65 ± 2.62^d	48.91 ± 2.28^d	145971.34 ± 12907.08^d	225.09^d
	PVC leachates	0	384.05 ± 4.95^a	197.18 ± 5.91^a	98221.63 ± 2188.37^a	121.28 ± 4.43^a	77.02 ± 3.17^a	510756.67 ± 28826.14^a	53.64^a
		0.3	359.70 ± 5.53^b	185.69 ± 6.42^b	89500.78 ± 3371.78^b	108.52 ± 1.86^b	62.99 ± 4.86^b	330556.99 ± 20871.85^b	102.24^b
		1	281.46 ± 6.41^c	173.55 ± 6.50^c	68365.14 ± 3846.81^c	80.27 ± 3.26^c	57.15 ± 2.73^c	170330.64 ± 17357.91^c	159.93c^c
		3	264.68 ± 6.80^d	156.74 ± 4.34^d	57675.60 ± 7533.39^d	71.52 ± 2.56^d	49.18 ± 2.35^d	115314.22 ± 9602.51^d	211.12^d
		10	250.18 ± 6.73^e	129.51 ± 6.72^e	45427.07 ± 3213.89^e	60.00 ± 5.00^e	44.46 ± 2.62^d	75459.89 ± 15128.31^e	255.14^e

Note:Data are mean ± SD. The abbreviations in this table represent body length (BL), body weight (BW), horizontal stomach diameter (S1), vertical stomach diameter (S2), stomach volume (SV) and index of microplastic impact (IMPI). Growth parameters between different concentrations were all analyzed by one-way ANOVA, with different letters indicating statistical differences ($p < 0.05$).

of 67.82%, 83.67%, 50.98% (at 24 hpf) and 65.15%, 65.68% and 46.25% (at 48 hpf) of the normal developed individuals. Generally, formation of digestive tract in sea cucumber larvae starts around 26 hpf (Wang et al., 2017). We found that PVC microspheres and leachates significantly delayed the development of digestive system starting from the concentration of 1 mg L^{-1} of PVC-microspheres ($p < 0.05$) and 0.3 mg $L^{-1}$of PVC-leachates ($p < 0.05$), respectively, which is similar to the results of body shape index. At the highest concentration of PVC-microspheres and PVC-leachates (10 mg L^{-1}), the S1, S2 and SV of larvae account a percentage of 67.32%, 63.50%, 28.5% (PVC-microspheres) and 49.47%, 57.73% and 14.77% (PVC-leachates) of the normal developed individuals at 48 hpf (Fig. 1 and Table 3).

3.2.2. Morphological classification

To evaluate the developmental toxicity of PVC microspheres and leachates more precisely, a morphological deviation of embryos and larvae from the wildtype (Table 1 and Fig. 2) was used to establish an index of microplastic impact (IMPI). To better describe the specific morphological changes of embryos exposed to different treatments, we captured the representative embryo images with the highest percentage of classification in each treatment group (Fig. 4). At 24 hpf, the embryos of control group show a phenotype corresponding to late gastrula stage for *A. japonicus* (classified as 0 level): the blastula is regular oval in shape and the mesenchymal cells are normally distributed around the blastodium (seen in Fig. 4A and A′). No obvious morphology changes of mesenchymal blastula were observed after exposure to 0.3 mg L^{-1} PVC-microspheres compared to the control embryos (Fig. 4B, also classified as 0 level). Exposure to intermediate concentrations of PVC-microspheres and PVC-leachates (Fig. 4C, D, B ′and C′) induced a different degree of delay in the elongation of the archenteron as compared to the control embryos (classified as 1 level), yet the embryo size became smaller with the increase of exposure concentration. High concentration of PVC-microspheres and PVC-leachates can further exacerbate deformities of *A. japonicus* embryos with weird shapes (classified as 3 level), and the embryo even showed ectodermal rupture and tissue dispersed at the highest concentration of PVC-microspheres and PVC-leachates (Fig. 4E, D′, E′). At 48 hpf, the health larvae have developed a complete digestive system with the formation and differentiation of four arms, which showed a typical phenotype corresponding to early auricularia stage for *A. japonicus* (classified as 0 level, seen in Fig. 4F and F′). As for exposure to 1 mg L^{-1} of PVC-microspheres and 0.3 mg L^{-1} of PVC-leachates, although the larvae were mostly classified into the same class of wild-type (0 level), these larvae showed smaller stomach volume and developmental delay (Fig. 4H and G′). The majority of larva exposed to 3 mg L^{-1} PVC microspheres were defined as Class 1 (Fig. 4I), with four arms poorly developed. However, exposure to a lower concentration of PVC-leachates at 1 mg L^{-1} (Fig. 4H') resulted a higher grade level in sea cucumber larva (classified as 2 level). Similar to the gastrula stage, highest concentration of PVC exposure (10 mg L^{-1}) also leads a significant damage to the larvae with obvious broken tissues (Fig. 4J and J'). It is worth mentioning that the endocytosis of PVC microspheres was occasionally detected in the larvae of *A. japonicus* (Fig. S1).

As shown in Fig. 5, the embryos exposed to PVC microspheres and leachates at lower concentrations (less than 1 mg L^{-1}) at 24 hpf were classed in class 0 and 1, assigning an IMPI less than 100. At the highest tested concentration (10 mg L^{-1} of PVC-microspheres and PVC-leachates), the percentages of class 3 were 36.39% and 49.40%, assigning an IMPI of 178.13 and 219.60. At 48 hpf, a similar trend was also observed that the percentage of anomalous embryos and larvae with higher categories increased with the increasing exposure concentrations and duration, and the leachates have severe developmental toxicity than the microspheres.

3.3. Phthalate composition analysis

Chemical analyses performing on highest concentration of PVC microspheres and leachates (10 mg L^{-1}) quantified concentrations of 16 kinds of phthalates (PAEs) (seen in Table 4). Five kinds of phthalates were detected in the PVC microspheres and leachates, including DMP, DEP, DIBP, DBP and DEHP. All concentrations of the PAEs detected in the leachates were higher than that those in PVC microspheres and control groups, and highest phthalate concentrations reached 1166 mg L^{-1} in DIBP of PVC-leachates, which is 3.59-fold and 9.97-fold higher than those in 48 h PVC- microspheres and control group.

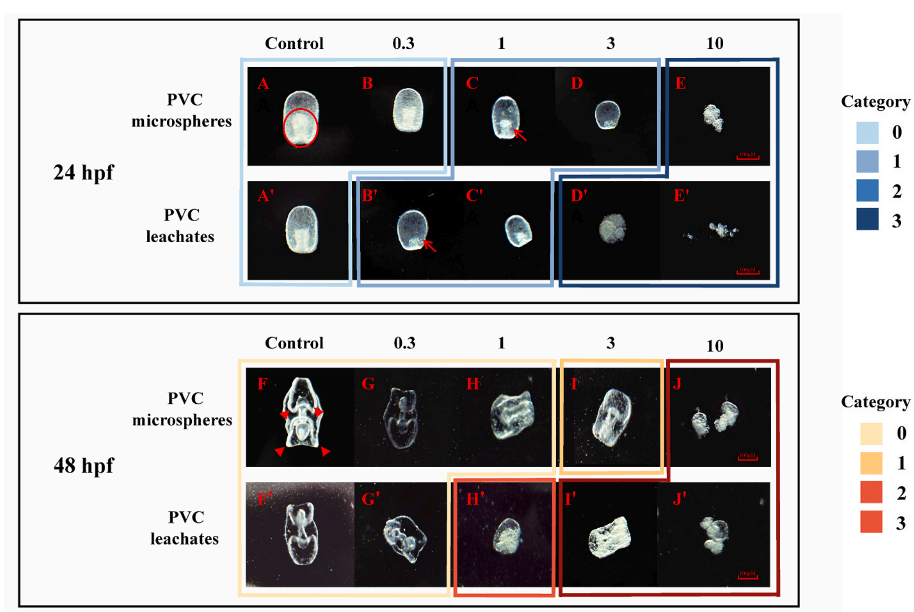

Fig. 4. Representative graded phenotypes of embryos and larvae in *A. japonicus* were compared at different treatments.
The circle represents embryo's endoderm. The arrows represent random distribution of mesenchymal cells. The triangles mark the arms of early auricularia stage larval.

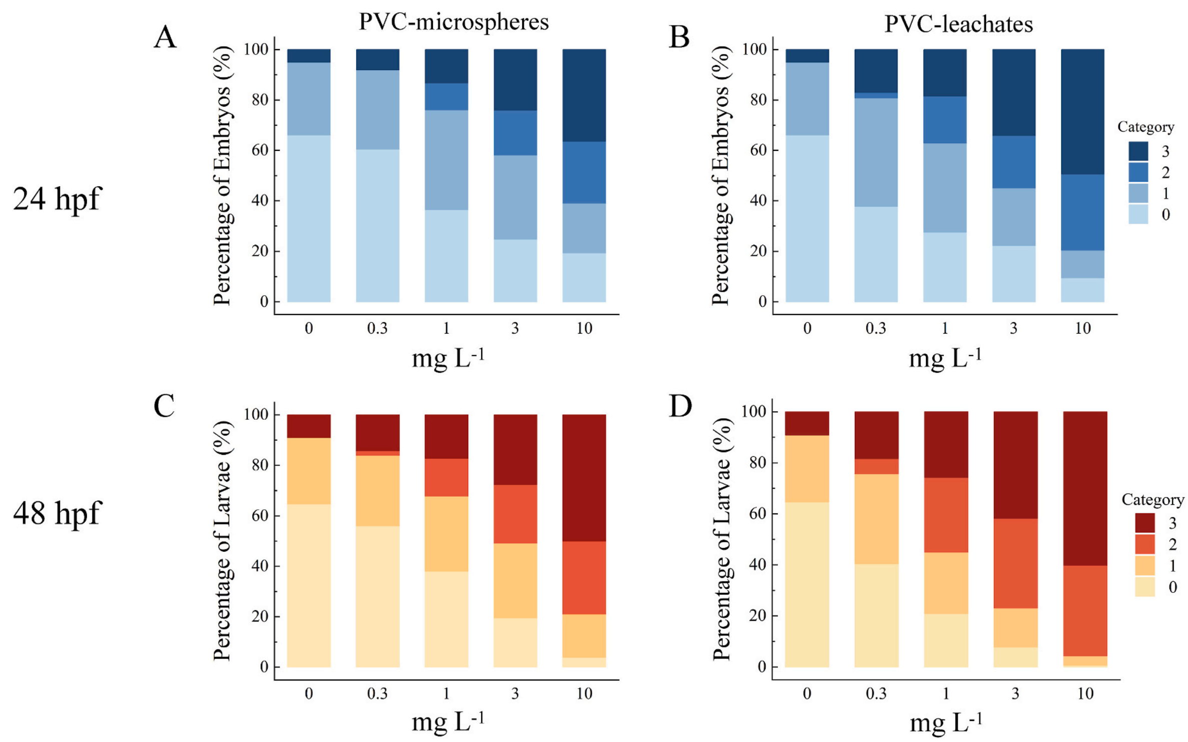

Fig. 5. Percentage of four categories in *A. japonicus* embryos and larvae after PVC exposure according to embryo development classification method. A. Percentage of four categories embryos at 24 hpf under different concentrations of PVC-microspheres. B. Percentage of four categories embryos at 24 hpf under different concentrations of PVC-leachates. C. Percentage of four categories larvae at 48 hpf under different concentrations of PVC-microspheres. D. Percentage of four categories larvae at 48 hpf under different concentrations of PVC-leachates.

Table 4
Concentrations of 16 kinds of PAEs detected in PVC microspheres and leachates.

PAEs (ng L^{-1})	PVC-microspheres (10 mg L^{-1})		PVC-leachates (10 mg L^{-1})	Control	Detection Limit	Recovery Rate(%)
Treat time	24h	48h	72h	–	–	–
DMP	330	383	394	149	50	103%
DEP	65	92	81	30	50	104%
DIBP	149	325	1166	117	50	107%
DBP	260	354	1070	268	50	114%
DEHP	155	155	159	102	50	69%
DPRP	<50	<50	<50	<50	50	105%
DiDP	<50	<50	<50	<50	50	51%
DPP	<50	<50	<50	<50	50	70%
DHXP	<50	<50	<50	<50	50	76%
DHPP	<50	<50	<50	<50	50	68%
BBP	<50	<50	<50	<50	50	86%
DCHP	<50	<50	<50	<50	50	62%
DnOP	<50	<50	<50	<50	50	47%
DiNp	<50	<50	<50	<50	100	45%
DDCP	<50	<50	<50	<50	100	50%
DUP	<50	<50	<50	<50	100	61%

Note: The sample plus the scalar was 500 ng L^{-1}. The abbreviations as follows: DMP: Dimethyl phthalate; DEP: Diethyl phthalate; DIBP: Diisobutyl phthalate; DBP: Dibutyl phthalate; DEHP: Di (2-ethyl) hexyl phthalate; DPRP: Di-propyl phthalate; DiDP: Di-iso-decyl phthalate; DPP: Diphenyl ortho-phthalate; DHXP: Dihexyl phthalate; DHPP: Di-heptyl phthalate; BBP: Butyl benzyl phthalate; DCHP: Dicyclohexyl phthalate; DnOP: Dioctyl phthalate; DiNp: Diisodecyl phthalate; DDCP: Di-decyl phthalate; DUP: Phthalic Acid.

4. Discussion

Although the deleterious effects of MPs on marine organisms have been widely studied, the majority of research is mainly focused on the effects of direct interactions of organisms with MPs, such as ingestion (Wright et al., 2013; Gall and Thompson, 2015; Mohsen et al., 2021). In recent years, a growing body of literature has demonstrated that the toxicological consequences of MPs may stem from a combination of physical and chemical toxicity (Browne et al., 2013; Teuten et al., 2009; GESAMP, 2015). However, when compared to the physical damage caused by direct exposure to MPs, the chemical hazards associated with indirect exposure to MPs appear to be more insidious and detrimental to marine biota (Capolupo et al., 2020; Schiavo et al., 2020). In order to compare the physical and chemical toxicity of MPs (Nobre et al., 2015), the comparative experiments were performed by setting PVC microspheres and leachates exposures respectively in *A. japonicus* embryos and larvae in this study.

The results revealed that sea cucumber zygotes exposed to virgin PVC microspheres and their leachates showed a significant toxic effect on their embryo development and larval growth ($p < 0.05$). Notably, the embryotoxicity of polyvinyl chloride (PVC) microspheres and their leachates on sea cucumbers exhibits a dose-dependent and time-dependent pattern, consistent with previous studies on sea urchins. Similarly, polystyrene (PS) and polymethyl methacrylate (PMMA) microplastics exposures resulted in significant concentration-related increase of developmental defects in plutei of sea urchin *Sphaerechinus granularis* (Trifuoggi et al., 2019). This suggests that phenotypical aberrations in embryos and larvae are linked to the concentration and duration of MP exposure (Manzo and Schiavo, 2022). However, Martínez-Gómez et al. (2017) have reported that a higher embryonic abnormality of sea urchin (*Paracentrotus lividus*) was observed exposed to 10^3 PS microspheres/mL FSW than to 10^5 PS microspheres/mL FSW. They attributed this phenomenon to the aggregates of PS microspheres in the highest concentrations, which may limit their potential toxicity.

Furthermore, the leachates of PVC microspheres present greater embryotoxicity and larval toxicity than the microspheres in our study. A variety of aquatic organisms have been adversely affected by leachates from plastic products in previous studies (Lithner et al., 2012; Bejgarn et al., 2015; Li et al., 2015; Gandara e Silva et al., 2016; Magni et al., 2018; Oliviero et al., 2019). And the leachates of MPs have usually been shown to be more toxic than the plastic pellets in aquatic environments (Bejgarn et al., 2015; Li et al., 2015). For example, PE and PVC leachates both have a significant embryotoxic effect on the sea urchin *P. lividus* (Beiras et al., 2019; Oliviero et al., 2019), and raw PE and PVC microplastic resins did not provoke larval toxicity in *P. lividus* (Beiras et al., 2021). Piccardo et al. (2020) also reported *P. lividus* larval growth was less evidently affected by the exposure to PET microplastic suspensions in different sizes than to the relative leachates. Actually, the toxicity of plastic leachates is mainly originated from the composition of plastics and the entered additives during the manufacturing, hence the toxicity of leachates varies with plastics type and plastic processing. Lithner et al. (2012) compared the toxicity of leachates from five different plastic products to *Daphnia magna*, and found that plasticized PVC leachates were among the most toxic but leachates from polypropylene, ABS, and rigid PVC products did not show toxicity. Moreover, various contaminants accumulating on plastic surfaces were also responsible for the high toxicity of plastic leachates, including persistent organic pollutants (POP), pesticides or dichlorodiphenyl-trichloroethane (DDT) (Ogata et al., 2009; Fries and Zarfl, 2012; Kedzierski et al., 2018). Such conditions are generally found in weathered plastics in realistic marine environment. For example, a study of Silva et al. (2016) indicated that the toxicity of the leachate from beached pellets was much higher than that of virgin pellets to brown mussels *Perna*. Hence, the toxicity of weathered plastic leachates may be a complex result of a combination of multiple factors, including types of polymers, species sensitivity, weathering conditions, and adsorbed contaminants (Endo et al., 2005).

In the present study, the embryotoxic effect on *A. japonicus* zygotes was assessed using virgin plasticized PVC pellets and leachates. PVC is a widely utilized material due to its advantageous technical properties. To attain the desired functionality, high concentrations of additives, predominantly phthalates, comprising more than 40% by weight of plasticizers, are frequently employed (Andrade et al., 2021). However, phthalates are easily leached out from the plastic particles due to their lack of covalent bound to the plastic molecules (Liu et al., 2022), which have been proven to be toxic for various aquatic organisms (Mathieu--Denoncourt et al., 2016). As a result of industry policies on privacy, we are unable to disclose details about the types and concentrations of plastic additives added to the virgin PVC granules used in the present study. Hence we detected 16 types of PAEs in the highest concentration of PVC microspheres and leachates (10 mg L^{-1}), and the results indicated that DIBP and DBP concentrations in the PVC-leachates were much higher than those in PVC- microspheres and control group. DIBP is a male reproductive toxicant and is identified as a presumed human health hazard by the National Academy of Sciences (NAS) (CPSC, 2014; NAS, 2017). In male rats and mice, gestational exposure to DIBP led to decreased testosterone levels and adverse effects on sperm or testicular histology. Additionally, existing research suggests that DIBP causes developmental toxicity, which increases post-implantation loss and decreases pre- and postnatal growth (Saillenfait et al., 2006, 2008, 2017; Wang et al., 2017). DBP, as a priority controlled toxic pollutant listed by the United States Environmental Protection Agency (USEPA) and China (Jiang et al., 2016), has been reported to induce potential endocrine disruption and developmental toxicity in shellfish, fish, and amphibian (Brannen et al., 2010; Zhou et al., 2011; Manikkam et al., 2013; Gardner et al., 2016; Xu and Gye, 2018). Notably, DBP can have strong teratogenic effects in the early stages of aquatic organisms. In zebrafish embryos, DBP affects the formation of the dorsal-ventral axis and results in abnormal morphology (Zarragoitia et al., 2006; Fairbairn et al., 2012). And for tadpoles of *Xenopus laevis*, exposure to DBP at 10.0 mg L^{-1} frequently resulted in ventral blisters, abnormal gut coiling and head malformation, hence DBP is considered to the most teratogenic phthalate ester plasticizer for amphibian embryos, with the highest teratogenic index among its counterparts (Xu et al., 2022). Based on the results above, we suggest that the high toxicity of leachate to sea cucumber embryos might have been mainly derived from phthalate ester plasticizers, especially DIBP and DBP. However, there is lack of definite evidence correlating the embryonic deformities of sea cucumbers and PAEs exposure in the present study. Future work is needed to further elucidate the teratogenic mechanism of PAEs exposure during the embryo development of *A. japonicus*.

In this study, we observed abnormal phenotypes of the embryos and larvae of *A. japonicus* exposure to virgin PVC microspheres and their leachates. Numerous previous studies have confirmed that plastic particles and leachates can cause various levels of developmental toxicity in marine species. For example, sea urchin *P. lividus* exposed to PVC and leachates showed reduced larval length, a block of development and severe developmental abnormalities at both embryonic and larval stages (Oliviero et al., 2019; Rendell-Bhatti et al., 2021). Another study of Silva et al. (2016) has indicated the leachate from beached pellets completely impaired the development of brown mussel *P. perna* embryos (*Perna*), as all the individuals were dead or abnormal. In general, the embryonic development of sea cucumber *A. japonicus* reaches the gastrula stage at 24 hpf with a formation of gastral primordia from the plant pole, up to half height of the germ body. Later, the top cells of gastral primordia generate mesenchymal cells (PMCs) with secreting a biomineral structure (spicules), which is closely associated with late skeletal development of *A. japonicus* larva (Zhu, 2009). However, if symmetric PMC clusters do not form properly around the blastopore at this stage, spiculogenesis will not be initiated, and skeletal formation will be affected (Decker and Lennarz, 1988). Moreover, exposure to PVC microspheres and their leachates at low concentrations (less than 1 mg L^{-1}) can lead to delayed gastral primordia development and misarranged PMCs in the ventral region of the blastocyste in *A. japonicus* at 24 hpf, which is likely indicative of major skeletal developmental defects in the late juvenile stage. Similar observations have been reported by Rendell-Bhatti et al. (2021) that leachates from PVC, biobeads, and nurdles can cause defects in skeletogenesis of *P. lividus* larvae, resulting in the absence of skeletal formation, particularly severe in exposure to PVC leachates. Presumably, PCBs, PAHs and heavy metals leaching from plastic materials are likely to be responsible for these skeletal malformations. As Ju et al. (2012) found that high concentrations of Aroclor1254 (PCB) resulted in skeletal morphological abnormalities in zebrafish embryos, and sea urchins treated with PAHs and $NiCl_2$, showed suppressed spicule formation or altered dorsoventral axis of ectodermal cells (Hardin et al., 1992; Suzuki et al., 2015). The digestive tract of sea cucumber larvae begins to form about 26 hpf and a relatively complete digestive system has been formed at 48 hpf, which marks sea cucumber enters into the early auricular larva stage (Wang et al., 2017). At this time, the stomach expansion can be fed through the contraction of the pharynx and esophagus, and MPs found in the stomach of *A. japonicus* larva implies the risk of MPs internalization occurs much earlier than the previous reported studies (Mohsen et al., 2019, 2021). In the internalization process of MPs, particle size is a crucial factor to consider since zooplankton ingest MPs that are the size of their prey (Desforges et al., 2015). For example, Oliviero et al. (2019) demonstrated that *P. lividus* embryos could ingest plastic particles smaller than 20 μm. Moreover, accumulation of MPs in the digestive tract can produce false satiety that further affects the feeding rate, might resulting in growth retardation (Oliviero et al., 2019). Since marine larvae are susceptible to predation, which may serve as a pathway for microplastics (MPs) to enter higher trophic level organisms. The transfer of MPs between trophic levels has already been demonstrated to occur among adult marine invertebrates (Farrell and Nelson, 2013).

5. Conclusion

This study shows that sea cucumber *A. japonicus* embryo-larval development is sensitive to both microplastic PVC particles and their leachates, and the toxicity is directly dependent on the concentration of plastic particles and the exposure duration. We also found that the leachates of PVC present a greater toxicity than the microspheres themselves, suggesting microplastic debris may be harmful to sea cucumber even they are not direct exposed or ingested by organisms. Furthermore, the high concentrations of PAEs detected in the PVC-leachates explain that the high toxicity of plastic leachate is likely due to phthalates additives. Our study will provide valuable information for clarifying the toxicity of MPs and their leachates on marine invertebrates, leading to better ecological risk assessment of MPs.

Author contribution statement

Haona Wang: Material preparation, data collection and analysis, Writing- Original draft preparation; Hui Liu: Methodology, Visualization, Software; Yanying Zhang: Writing- Reviewing and Editing; Lijie Zhang: Material preparation; Qing Wang: Conceptualization, Foundation, Writing- Reviewing and Editing. Ye Zhao: Conceptualization, Foundation, Writing- Reviewing and Editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ye Zhao reports financial support was provided by National Natural Science Foundation of China.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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